Knowledge and Information Systems (2024) 66:4405–4436
https://doi.org/10.1007/s10115-024-02122-6

REVIEW

Machine learning and deep learning models for human activity recognition in security and surveillance: a review

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Received: 1 December 2023 / Revised: 26 March 2024 / Accepted: 13 April 2024 /
Published online: 4 June 2024

Abstract
Human activity recognition (HAR) has received the significant attention in the field of security and surveillance due to its high potential for real-time monitoring, identifying the abnormal activities and situational awareness. HAR is able to identify the abnormal activity or behaviour patterns, which may indicate potential security risks. HAR system attempts to automatically provide the information and classification regarding activities performed in the environment by learning the data captured through sensor or video stream. The overview of existing research work in the security and surveillance area, which includes traditional, machine learning (ML) and deep learning (DL) algorithms applicable to field, is presented. The comparative analysis of different HAR techniques based on features, input source, public data sets is presented for quick understanding, and it focuses on the recent trends in HAR field. This review paper provides guidelines for the selection of appropriate algorithm, data set, performance metrics when evaluating HAR systems in the context of security and surveillance. Overall, this review aims to provide a comprehensive understanding of HAR in the field of security and surveillance and to serve as a basis for further research and development.

Keywords Human activity recognition · Machine learning · Deep learning · Security · Surveillance

1 Introduction

Human activity recognition (HAR) system gives labelling to human actions and classify them into the respective category. Significant improvements have been made in HAR for security and surveillance in recent years due to the increasing availability of sensor technologies, high-resolution video cameras and the development of sophisticated ML and DL algorithms. The basic HAR system includes data gathering, feature extraction either handcrafted features or

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automatically extracted features, feature selection and classification. HAR is in high demand as it has numerous applications available in different areas. HAR can be beneficial for number of applications including industry, smart city, intelligent surveillance at public places like mall, stations, abnormal activity detection, medical, sports analytics, fitness, daily-assisted living, pedestrian detection, and so on [1]. An ideal human action recognition system should be able to adapt to differences within a single action class and distinguish between variations within and between distinct action classes. The data required for performing HAR are either sensor based or vision based; sensor data set means information collected from different sensors like accelerometer, magnetometer, gyroscope, RFID, etc. [2], and vision data set includes information derived from CCTV camera, RGB depth camera like Kinect sensor. Pre-processing is performed as initial step. In case of sensor data, pre-processing techniques such as data cleaning, handling missing values, dimensionality reduction, data transformation are used. For vision data, usually image processing tools are applied for removing digital noise and perform data augmentation. By using this processed received information, human activity is detected by traditional, ML or DL based approach. Extracting features in order to predict the activity is the main and challenging task. Various traditional representative feature descriptors are Haar, histogram of oriented gradient (HOG), local binary pattern (LBP) [3, 4]. Traditional methods are based on global features of an image and are only suitable for individual action detection. The feature extraction with the help of traditional methods is computationally intensive. These issues can be handled more effectively with the help of ML-DL-based algorithms. Lightweight architectures such as MobileNet, SqueezeNet, and Tiny-YOLO are well-suited for implementation on resource-constrained devices for human activity recognition. In addition, authors [5] proposed posture recognition system using mobilenetV2 and LSTM to extract important features. On other hand, authors [6] proposed Tiny YoLo-based real-time human detection for video surveillance at the edge to achieve a good trade-off between accuracy and processing time. The survey [7] highlights the multidisciplinary aspect of HAR, covering application domains, activity types, task complexity, benchmark data sets, and techniques. This research compares current HAR approaches and discusses common data sets. The proposed method [8] discusses the integration of biosensor data and multimodal deep learning techniques for decoding human locomotion patterns in the context of the Internet of Healthcare Things (IoHT). The authors focus on deep human motion detection and multi-feature analysis as essential components for developing smart healthcare learning tools, contributing to more effective and personalized health assessments [9]. HAR has achieved significant importance due to its wide range of applications and corresponding features in variety of domains and are listed in Fig. 1.

In contrast to other applications, HAR in security and surveillance carries a special significance because of the seriousness of its goals and the potential consequences of failure. Compared to other domains like health care, sports, or human–computer interaction, HAR function in security and surveillance has a direct impact on public safety, property protection, and crime prevention.

Figure 2 shows the characteristics and challenges of HAR in security and surveillance domain. The vision-based data set specifically CCTV footage is important for HAR in security and surveillance. Feature extraction is a critical characteristic of HAR for security and surveillance, as it transforms raw data into discriminative and context-aware representations, enabling accurate and real-time detection of human activities. Getting the real-time data is crucial, and also there is lack of sufficient data [10, 11]. In addition to recognizing predefined activities, HAR systems can also identify anomalies or suspicious behaviours that may indicate a security threat. Real-time processing is crucial for security and surveillance applications. HAR systems often operate in real time, allowing immediate detection and response

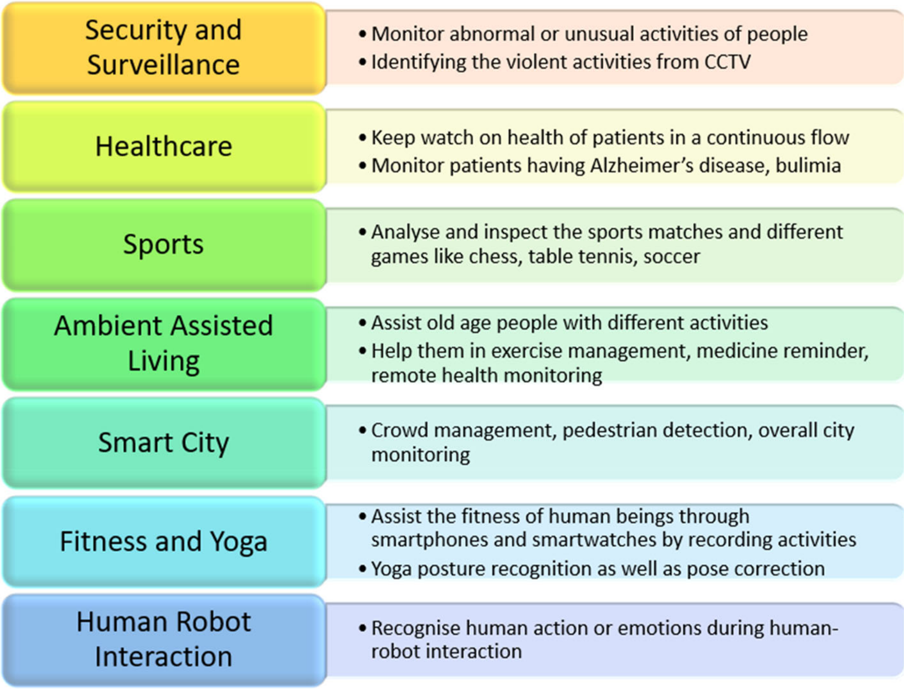

Fig. 1 Contribution of HAR in applications

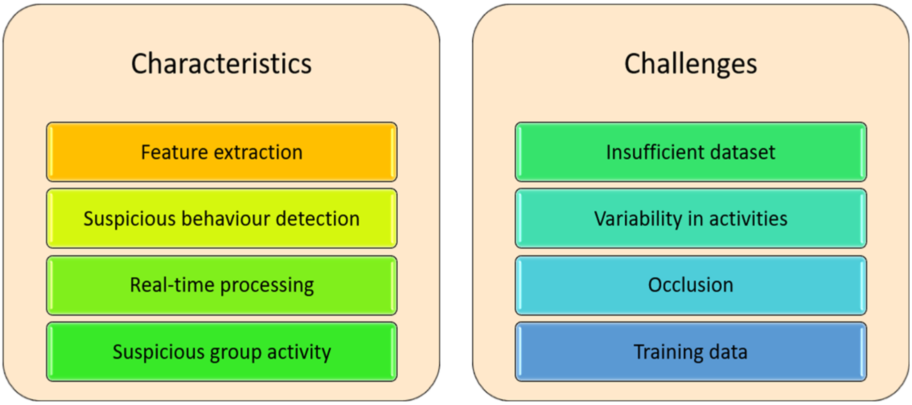

Fig. 2 Characteristics and challenges of HAR in security- surveillance

to suspicious or anomalous activities. Human activities can vary significantly in terms of appearance, context, and execution. Recognizing activities accurately in diverse condition is a major challenge. Overlapping objects leads to partial or full occlusion, and it creates difficulty in predicting the activity. Gathering and labelling diverse and representative training data for HAR in security and surveillance models are challenging. It becomes difficult to recognise the activity when more people are there in group. Human interaction can lead to misclassification.

1.1 Motivation

The main purpose of the review is to present a current status of HAR in security and surveillance. It has ability to detect threat, deliver real-time monitoring, optimize resource allocation, facilitate investigations, enable proactive responses, provide scalability and coverage, and integrate with other security systems. Global Market Insights, Inc.1 forecasts that the market for behaviour analytics would reach over $3.5 billion by 2024, and it has many research applications [12, 13]. HAR usage contributes to the overall effectiveness and efficiency of security operations, promoting safety and security in various environments.

1.2 Principle of HAR

The general process of HAR is illustrated in Fig. 3. The input data captured will be either sensor-based (smartwatches, smartphones, etc.) or vision-based (CCTV, RGB data) [14]. Several data sets are publicly available online for processing. The data must go through the pre-processing stage for noise removal, data transformation, and data cleaning. Vision-based data need different filters and sensor-based data require domain transformation techniques for data cleaning. The pre-processed data then go through various steps such as feature extraction, feature selection, and ML or DL classifier [15, 16]. System performance can be measured using various performance metrics, while training the model based on application requirements. In order to improve the efficiency of the network, it is necessary to minimize objective function. In this aspect, the optimization algorithm plays an important role [17]. Various optimizers are available for getting optimal results. This allows human activities to be classified into a particular class for maximum accuracy and improved model performance.

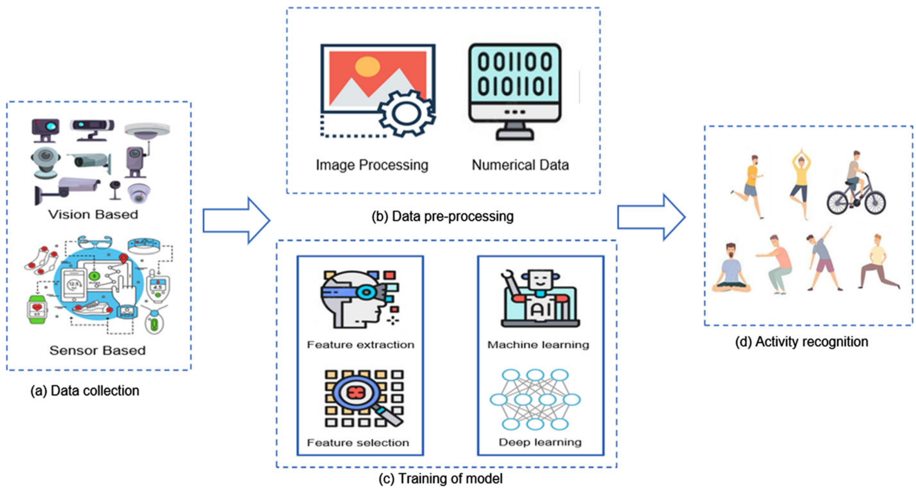

Fig. 3 A general approach to human activity recognition

ML algorithms have been applied for identifying and analysing the activities for security and surveillance systems. ML algorithms are mainly classified as supervised algorithm, unsupervised algorithm and reinforcement algorithms. Numerous ML algorithms such as support vector machine (SVM) K-nearest neighbour (K-NN), multi-layer perceptron (MLP), naive Bayes classifier [18] are applicable for performing HAR. For the supervised machine learning approach, the labelled data set requirement is there. The supervised learning suffers from generating training data set and extending activities. Some work in semi-supervised manner to overcome this problem is also presented. The unsupervised clustering ML algorithm is applied to tackle the issue of supervised learning method. The feature extraction in ML needs deep knowledge of application domain or human experience. These algorithms differentiate between various activities and assist in identifying potential threats, maintaining security.

However, with the advent of deep learning, researchers are increasingly turning to neural network architectures to tackle the complexities of HAR. Before the widespread adoption of DL, traditional ML algorithms were extensively used for image classification, often relying on manual feature extraction. Unlike ML, DL extracts optimal features automatically. In DL, feature extraction is easy and requires repeated learning from itself. It reduces human efforts, feature engineering, need for expert knowledge. It enhances activity recognition accuracy of security and surveillance system. Activity is nothing more than the sequence of actions that follow specific pattern. Deep models learn the pattern and then predict the output. However, it requires large data set for operation in order to achieve good results. Various DL algorithms like convolutional neural network (CNN), restricted Boltzmann machine (RBM), recurrent neural network (RNN), long short-term memory (LSTM) [19], deep belief network (DBN) [20], deep neural network (DNN), and various hybrid deep models (combination of more than one deep model) [21] are available for human activity classification. CNN and RNN are most popular deep models. CNN is generally considered as a good for image data. It is scale-invariant and recognizes both spatial and temporal dependencies. RNN performs well on time series data because it has recurring connections and passes the previous information to next layer. RNN has disadvantage of gradient exploding or vanishing problems. The RNN versions like LSTM, stacked LSTM, bi-directional LSTM overcome this drawback as they have long-term time dependency. The DNNs are somehow different from traditional classifiers [22]. The other state-of-the-art technologies based on DNNs such as region-based convolutional neural networks (R-CNNs), Fast R-CNN, Faster R-CNN, and single-shot detector (SSD) have achieved success in object classification [15]. Hybrid models such as combinations of CNN and LSTM, 1-D-CNN and SVM, SIFT and SVM, CNN and RNN-GRU are also generated to evaluate HAR performance. The rest of the paper is organized as follows. Section 2 presents background information, available research methods, algorithms for HAR, and Sect. 3 provides the key points for the selection of data set. Section 4 provides the significance of work flow in HAR. The article's final section discusses the summary and suggestions for future research.

2 Background

HAR enhances the capabilities of surveillance systems and enables proactive measures to ensure safety and security. Many surveys in the area of security and surveillance focus on different approaches of activity recognition. Many reviews on HAR in field of security and surveillance have been conducted [23], the fact that HAR is still in its development leads to the introduction of many new concepts. HAR mostly uses either sensor-based data or vision-based data as an input source for identification. This review paper aims to provide available literature and information for both types of data sets in order to gain insights for the security and surveillance application. There are different levels for HAR depending on increasing complexity. The simplest is single person action recognition such as gesture recognition, posture recognition [24], fall detection. Some are action recognition like running, walking, etc. The complexity increases for interaction-based activities like human–object interaction and human–robot interaction [25], while the rest are motion based like tracking, motion detection and people counting. For the security and surveillance application, the complexity of activities is higher always as it involves suspicious activity identification, group activity detection, abnormal activity detection and analysis. "Deep Learning approaches for Human Activity Recognition in Video Surveillance—a Survey," [26] provides a comprehensive overview of HAR in video surveillance. It covers a wide range of topics, including feature extraction methods, classification algorithms, multi-modal approaches, and benchmark data sets used in HAR research for security and surveillance. Chen et al. [27] focuses on deep learning-based HAR methods specifically applied to video surveillance scenarios. "A Survey on Human Activity Recognition Using Wearable Sensors" provides an in-depth analysis of HAR using wearable sensors. It covers various sensor modalities, feature extraction techniques, and classification algorithms employed in wearable-based HAR systems for security and surveillance applications. Khushboo Banjarey [28] presents survey on vision-based approach for activity detection. Deshpande et al. presented a vision system for detection of human in videos using support vector machine. To protect privacy, Wi-Fi channel state information is used as a source for HAR.

2.1 Methods of HAR

The HAR methods presented in this section are categorized into the subsections—traditional method, ML method, DL method and hybrid model method implemented so far in security and surveillance area. These methods need to be selected based upon the complexity of activities, data set size, time complexity, cost and all.

2.1.1 Traditional method

Traditional method is commonly useful for simple and individual actions. The authors [29] indicate an approach that uses background subtraction for human detection. Feature extraction and classification is carried out using HOG descriptor, SVM, respectively. This system is implemented for multiple human detection and video surveillance application, while [30] proposes a model that uses HOG for feature extraction, SVM for classification, but cluster segmentation approach is used to separate human from background or other objects in video frames. Sonal Kumar and Bhavani in [31] present HAR using median filtering for pre-processing, watershed segmentation technique. However, feature extraction is achieved by combining the three feature extractors HOG, colour and GiST. Congcong Liu, Jie Ying, and colleagues demonstrate how two separate classifiers—CNN and the Bayes classifier, are used to detect anomalous human behaviour [32].

2.1.2 Machine learning method

When the size of data set is small, ML method is preferrable in that case. The HAR system [33] mentioned is based on random forest (RF), which is compared with decision tree (DT) and artificial neural network (ANN). This HAR framework is based on single wearable device in centralized mode and it is proved that RF has outperformed. The RF binary classifier [3] is used to classify the abstract activity like static or dynamic activity and a low-pass Butterworth filter is applied to pre-process a signal. In [29], authors combined traditional method and ML method for classification, where SVM works as classifier for activity recognition. Random forests, gradient boosting machines are an ensemble technique that can be used for classification tasks. KNN is a simple algorithm that classifies an image based on the majority vote of its neighbours. The study explores the application of ML techniques to analyse and classify human activities using wearable sensors. The study emphasizes the potential of HAR in health care for early detection of abnormalities and enhancing the overall quality of patient monitoring [34].

2.1.3 Deep learning method

DL method is preferred when dealing with sequential and temporal data, as they capture long-range dependencies and temporal patterns, enabling accurate tracking and analysis of human activities over time. The most noteworthy modality aspects are automatically extracted using the attention mechanism and RNN, and then data are subsequently transformed to higher-level representation. Sample data are evenly sampled and classified into partitions using K-means clustering to train multi-modal classifiers. On the basis of dense connections and weighted feature aggregation, a network of convolutional LSTM for HAR is suggested. Training and tests were carried out using the benchmark data sets Opportunity and UniMiB-SHAR [37]. The main contribution of this work is the application of a dense connection module that has $L(L + 1)/2$ direct connections, while a traditional convolutional network has L connections for L layers. Sliding window length, kernel size, fusion node of dense connection module, and several convolutional layers are the hyperparameters that need to tune during the training of the network. While using CNN, the softmax function is used at the last layer which gives prediction probability for three moving activity classes (walking, walking upwards, walking downwards). Three hidden layers are used in this hybrid model, giving the best performance with high accuracy. As we increase the number of hidden layers, the complexity of the neural network increases, and performance decreases. Kalman filter is utilized to detect a target in a moving frame, and three features length–width ratio, entropy, and Hu invariant moment are extracted from the frame [35]. The ReLu activation function is applied for all hidden layers, and performance is measured for three extracted features by taking mean and variance. Kaixuan Chen, and Lina Yao et al. focus on three challenges in HAR: (i) unavailability of labelled data, (ii) labelling of data is costly, (iii) imbalanced classification and (iv) interpersonal variability and interclass similarity [36]. In some cases [38], HAR is carried out with LSTM-RNN network by using passive infrared (PIR), motion, float and pressure sensors to monitor daily activities. The work in [39] uses LSTM again and identify activities using triaxial accelerometers data. In [40] the authors compared LSTM, 1D CNN and hybrid model with joint diverse temporal learning on Kasteren data set. The focus is mainly on less represented activities. The authors [41] applied CNN-LSTM on UCI-HAR and iSPL data set while in [52] LSTM-CNN is applied on three data sets, namely UCI-HAR, WISDM and OPPORTUNITY. In this, fully connected layers are replaced by global average pooling (GAP) layer in order to reduce the model parameters [43]. This is followed by a fully connected layer, batch normalization and softmax function. To increase efficiency, small batches of data segment size are segmented during training and testing. The cross-entropy loss function is used to calculate the total error, and average accuracy is compared to other hybrid models.

2.1.4 Hybrid model method

Hybrid model is combination of traditional and ML algorithms or ML and DL algorithms. Some hybrid models combine the two DL models like CNN and LSTM. These hybrid models are popular as they give better results than single model approach. Authors [44] proposed combination of two techniques SIFT and SVM for activity detection. SIFT extracts local features from video, and SVM is applied as classifier for classification purpose. “Human action recognition using a hybrid deep learning heuristic” presents a combined SIFT and CNN model applied on two data sets which has extra-ordinary ability to extract features from both spatial and temporal dimension like 3D convolution. Hybrid model of CNN and GRU is proposed as an end-to-end model for performing automatic feature extraction and classification of the activities [45]. “A Hybrid Approach for Human Activity Recognition with Support Vector Machine and 1D Convolutional Neural Network”, in this paper hybrid model of SVM and 1D CNN effectively utilizes SVM and CNN for activity recognition using a waist mounted accelerometer and gyroscope sensor for getting higher accuracy [46]. The authors [47] proposed an automated method of using feature fusion and ML. The extracted features are obtained from HOG, Gabor and chromatic feature vectors. PCA is applied separately to each method in order to select important features. Huaijun Wang, Jing Zhao et al. [42] explained the HAR using a combination of CNN and LSTM, where CNN is used for feature extraction by using 3 convolutional layers. The feature map is classified with help of LSTM by using feature sequence fusion which depends on vector dimension size and improves the classification rate. All these methods used for HAR in security and surveillance field are equally important and involve different algorithms to perform activity detection. Traditional methods are complex to analyse and understand, but require less computational power. The selection of ML or DL method can be based on the size of data set and the application. For data-hungry applications, it is always beneficial to use DL algorithms. Hybrid methods takes advantage of ML and DL approach. Various algorithms available under all these methods are discussed in Sect. 2.2.

2.2 Algorithms for HAR

The number of traditional, ML, DL and hybrid algorithms that can be used for HAR are summarized and represented in Fig. 4. It gives the use of all algorithms for HAR is in detail manner. Different algorithms have different assumptions and knowledge bases depending on which accuracy will vary. Hybrid algorithms can also be implemented by a combination of two or more base algorithms to improve the network performance.

2.2.1 Traditional algorithms

These algorithms are grouped into texture metric and Fourier feature. Texture metrics describe the spatial arrangement and variation in pixel intensities or colours within an image and can be used for tasks such as texture classification, segmentation, and analysis. Fourier features capture frequency domain information using Fourier analysis. They are useful for

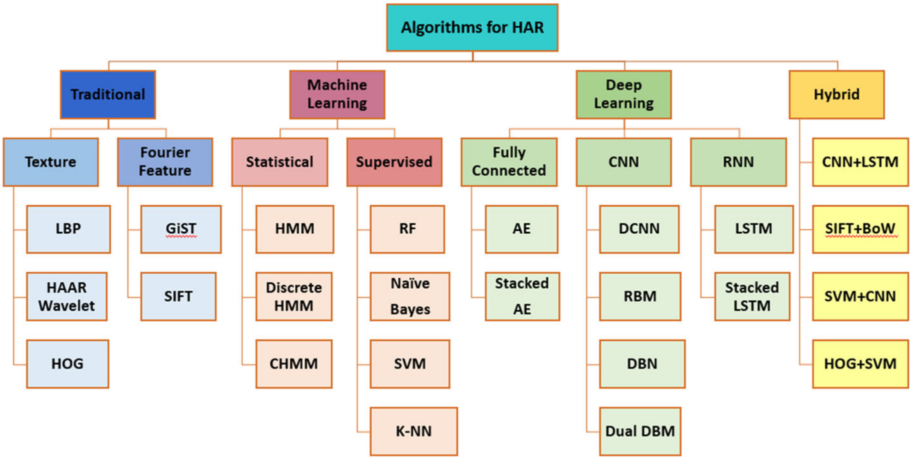

Fig. 4 Various traditional, ML, and DL algorithms for HAR

the detection and identification of activity, usually suitable for straightforward activities with limited computational resources.

1. **Local Binary Pattern (LBP):** LBP is a statistical texture descriptor helpful for feature analysis. Because it is a traditional method, it cannot be directly applied for action identification. For better performance, it can be combined with other models like SVM, HMM, and many others [55].
2. **Haar Wavelet**s: A 1D Haar filter is applied, which is nothing more than a simple filter for extracting features from data. The features are also simple (mean, standard deviation, correlation, and energy) and have low computation costs. The accuracy is calculated with help of the computation cost of features [56].

LBP: local binary pattern, HOG: histogram of oriented gradient, SIFT: scale invariant feature transform, HMM: hidden Markov model, RF: random forest, SVM: support vector machine, K-NN: K-nearest neighbour, AE: autoencoder, CNN: convolutional neural network, DCNN: deep convolutional neural network, RBM: restricted Boltzmann machine, DBN: deep belief network, RNN: recurrent neural network, LSTM: long short-term memory, BoW: bag of words.

3. **Histogram of Oriented Gradient (HOG):** It is commonly used for object detection. It selects useful information and discards the remaining part. It uses the direction of intensity of the gradients and edge directions. HOG calculates the change in gradient direction for a given specified portion. It is then followed by SVM or RF classifier for human detection and recognition [57].
4. **GiST**: GiST is a feature extraction technique based on the Gabor filter and convolution process. The Gabor filter is a texture analysis technique that considers the frequency and orientations. The convolution is carried out in the Fourier domain in order to perform computation. GiST filters the image into a number of visual feature channels such as colour, intensity, and orientation [56].
5. **Scale Invariant Feature Transform (SIFT):** Initially the pre-processing of the input image/frame is performed by removing noise, foreground, and background subtraction. SIFT is invariant to scale and rotation. The algorithm mainly performs 4 functions for

identifying features. Initially localize humans, determine a region of interest (ROI) for locating global feature key points, and the orientation of key points, and then reproduce them in high-dimensional order. The extracted global features apply to SVM for the classification purpose of actions [57].

For HAR tasks in security and surveillance, HoG and LBP are often preferred due to their ability to capture relevant spatial information and robustness to variations in lighting and texture. SIFT and GiST may have more limited applicability.

2.2.2 ML algorithms

ML algorithms are grouped into supervised and statistical algorithms. Supervised algorithms utilize labelled training data, unsupervised algorithms extract patterns and clusters from video data without prior labels and reinforcement algorithms make sequential decisions in response to human activities. They outperform traditional algorithms and provide good results for human activity recognition.

1. **Hidden Markov Model (HMM):** It is a statistical and probabilistic model based on Bayes theory. It consists of two stages. In the first stage, it learns the data set for each parameter and trains the model. In the second stage, it calculates the probability for each sequence class and recognizes the activity based on maximum probability [58]. It can perform on any length sequence and works easily for feature extraction as well as classification which is nothing but an advantage of using HMM [59].
2. **Discrete HMM:** D-HMM estimates the maximal posterior probability to predict the target.
3. **Continuous HMM:** C-HMM refers to Markov chain with a finite number of states and has a set of density functions representing the observations. It detects the sequential information from time sequence. The proposed system monitors everyday locomotion with body-worn inertial sensors and other electrophysiological sensors. It groups the HMMs into three body-related sensors placement, namely, active head, active mid-body, and active lower-body sensors [60].
4. **Random Forest (RF):** RF is a supervised machine learning algorithm based on the bagging ensemble technique, widely applicable for both classification and regression. It gives good results for activity classification. It forms a number of decision trees by splitting the training data set. It selects the samples randomly and the output is based on majority voting. Thus, it maintains high accuracy than other decision tree algorithms. RF algorithm is determined by the original data set and the number of decision tree classifiers K [61, 62].
5. **Naïve Bayes Classifier (NB)**: The moving target is detected with the help of a filter. The features such as entropy, Hu moment, and length–width ratio are extracted from the detected frame. These are applied to the Bayes classifier. The mean and variance are computed using these three characteristics and based on its Gaussian distribution function is adopted for classification [63]. Bayes algorithm is applied to find out the output with the largest posterior probability and use conditional probability for classification purposes.
6. **Support Vector Machine (SVM):** The first step is extraction of statistical features (mean, RMS, standard deviation, Index of dispersion, Skewness, etc.), followed by a dimensionality reduction technique using principal component analysis (PCA). Subsequently, low-dimensional data are fed to the SVM model. SVM is a simple and discriminative type of classifier. It is based on maximizing the distance between support vectors. SVM is an old but still, very popular supervised ML algorithm in today's date also [64]. It was

developed for binary classification, but also gives good results when classifying multiple classes.

7. **K-Nearest Neighbour (KNN):** It comes under the umbrella of a supervised ML algorithm. KNN has many applications in text mining, face recognition, and pattern recognition and also can be used in HAR. The value of K decides the number of neighbours which is dependent on a data set and finding the value of K is a bit crucial. It uses neighbours and calculates the Euclidean distance with which to add the new data point to the corresponding group. It has the potential to solve low accuracy problems [54].

SVM, RF, and HMM are commonly used in HAR for security and surveillance because of their capacity to handle complex data and capture temporal connections.

2.2.3 DL algorithms

DL algorithms are classified into fully connected, CNN, and RNN based on some advantages over machine learning algorithms. DL models work well for unstructured data, feature extraction is automatic, and self-learning ability through backpropagation.

1. **Autoencoder (AE):** Autoencoder is a neural network that produces an output that resembles input. It has an encoder, decoder, and hidden layer (which gives an abstract representation of data). An ensemble of autoencoders can be used for HAR. Each AE is dedicated to a single activity and trained for offline mode. When new data arrive for the online phase, it will reconstruct output, and classification is made with the help of minimum reconstruction error. Select the threshold as the standard deviation of the error [53].
2. **Convolutional neural network (CNN):** CNN is a more established DL model that has several application areas. Its main attraction is automatic feature extraction. Various CNN architectures like VGG, AlexNet, and GoogleNet with different convolutional layers and parameters are available. For activity recognition, CNN models need to be defined first with their specifications such as the number of convolutional layers, pooling layers, hyperparameters, batch size, and epoch. The architecture must be trained and tested on the data set for computing efficiency and accuracy of the model [54].
3. **Restricted Boltzmann machines (RBM):** It is a generative stochastic model and type of neural network based on probability distribution. It is generally used for classification, regression, collaborative filtering, and feature learning. The RBM is composed of two layers: a visible layer and a hidden layer.
4. **Deep Belief Network (DBN):** It is also a generative model that stacks multiple RBN layers. It can be used both supervised and unsupervised manner. It has the potential to perform both feature extraction and classification. DBNs can be greedily trained layer by layer and can be useful in HAR applications. DBN can first be trained in an unsupervised way and then can be proceeded to supervised tunning [20]. Since the deep network here does not perform during gradient descent, its DBN can give satisfactory output by un-supervised pre-training.
5. **Recurrent neural network (RNN):** RNN is a type of neural network with internal memory and is best suited for sequential data. Thus, it can perform well for human action recognition as human activities happen in a sequential pattern. Current input and past output are taken into account when making a decision. The dense layer identifies the applicable features. It features a cascading of layers with a fully connected layer at the output side [39]. But RNN can suffer from vanishing and exploding gradients issues. To handle this problem, there is an LSTM network.

6. **Long short-term memory (LSTM):** It is the next version of RNN. Compared to RNN, it has a good capability of handling long-term sequences. HAR is a time-series analysis, so LSTM may be best suited for this. The main component of the LSTM is the memory cell that has the input gate, output gate, and forget gate. As it is a variant of RNN, it provides a self-recurrent path. Time series data are taken as input and applied to the LSTM model. It computes the features and forms the feature vector. The classification can be done with the help of any multi-class classifier [40].

In context of HAR for security and surveillance, CNNs, RNNs, and LSTMs can be particularly useful depending on data.

2.2.4 Hybrid algorithms

A hybrid algorithm in the context of human activity recognition (HAR) refers to an approach that combines multiple techniques or models from different machine learning or data processing paradigms to enhance the accuracy, robustness, or efficiency of activity recognition. These algorithms merge the strengths of different methods to overcome the limitations of individual techniques.

1. **CNN + LSTM:** Features of CNN and LSTM [73] are combined to improve the classification accuracy and overall performance.
2. **SIFT + BoW**: It uses hybrid combination of traditional algorithms. BoW is used as feature descriptor and activity classification is attempted with the help of SIFT [74].
3. **SVM + CNN**: This uses combination of DL algorithm 1-D CNN for feature extraction and ML algorithm SVM in order to classify the human activities [75].
4. **HOG + SVM**: It extracts useful information from image by calculating the occurrence of gradient and followed by SVM [76] to perform the classification task.

CNN + LSTM is most widely used hybrid algorithm for security and surveillance.

The concise summary of all the ML, DL and hybrid algorithms explained in Sect. 2.2 is provided in Table 1 referring to existing literature. It provides HAR in various field with different approaches. The table gives information regarding the publication year, author, input source, feature extracted from source, description and algorithm-based technique employed for the identification of human activities.

It can be seen that from Table 1 user has to make correct choice depending upon feature and technique. The general approach of HAR for security and surveillance is illustrated in Fig. 3. The different state-of-the-art techniques and the existing algorithms applied in case of HAR for security and surveillance are explained in Sect. 2.2. The data set plays pivotal role in the development and deployment of the algorithms. The analysis of algorithms is evaluated on the data set. Since there are many publicly accessible data sets for the HAR, choosing the right data set for the security and surveillance is crucial because it affects the performance of the model. The selection of the HAR data set in security and surveillance is covered in the following section.

3 Selection of data set

The selection of data set plays crucial role for HAR in security and surveillance. The data derived from either visual or sensory input sources are stored in the form of data set. Data sets serve as a foundation for building accurate and robust recognition systems and enable

Table 1 Different techniques used in HAR

Year	Author	Input source	Feature	Description	Algorithm based technique
2022 [54]	Yair A. Andrade-Ambriz et al.	Short video	Spatiotemporal	Improvement in accuracy parameter	Take advantage of time motion features with spatial location
2022 [56]	Yale Hartmann et al.	Smartphone	Interpretable categorical high level	Feature extraction for classification	Use interpretable high-level features
2021 [83]	Wen Qi et al.	Smartphone and Depth data	3D joints, angular velocity, orientation	Complex activities and automatic labelling	Designed Hk-medoids algorithm and calibration approach
2021 [58]	Zhihua Feng et al.	–	Spatiotemporal	Identifying activities under complex background	Region matching tracking algorithm, integration of local and global features
2021 [59]	Ahmed Snoun et al.	Video	Human skeleton	To convert video into single image that summarizes human movements	Used the extracted skeletons from the video frames-dynamic skeleton, skeleton superposition, body articulations
2021 [52]	Saurabh Gupta	Wearable sensor	Temporal	Reduce stress on feature engineering and manual feature extraction	Hybrid deep neural network model implemented CNN-GRU
2021 [34]	Suvarna Nandyal et al.	ATM Video	Background subtraction	Safety aspect of ATM, detection of continuous and discontinuous objects	Implemented KLT algorithm and Kalman filter to detect small number of features
2021 [47]	Fathia G. Ibrahim Salem et al.	Video	Background Subtraction	To support smart surveillance for detecting abnormal behaviour	A sequential of procedures applied for suspicious activity

Table 1 (continued)

Year	Author	Input source	Feature	Description	Algorithm based technique
2020 [62]	Huaijun Wang et al.	Acceleration & Gyroscope	Motion	Recognition of specific activities and transition between two different activities	Feature fusion using CNN and LSTM
2019 [63]	Tahmina Zebin et al.	Waist mounted	Raw-time series	Memory and execution time requirements for mid-range smartphone	Reduction in size and execution time due to weight quantization
2019 [54]	Samundra Deep et al.	Video frames	Depth	To automatically extract features and reduce computational cost	Employed CNN and transfer learning
2018 [33]	Congcong Liu et al.	Image	Length–width ratio, entropy, Hu invariant moment	Robust features to environment	Entropy, Hu invariant moments, length–width ratio are used as the features
2018 [32]	K. P. Sanal Kumar et al.	Image	Spatiotemporal	An efficient technique to recognize the activities	Combined three features and used genetic algorithm to improve performance
2018 [54]	Antonio Bevilacqua et al.	Inertial sensor	Spatiotemporal	To address problem of activity recognition in context of exercise program	Different five sensors are used for 16 activities and getting good results
2018 [49]	Praveen M Dhulavvagol et al.	Video frames	Scale invariant	Identifying the actions of human from the given video	Developed robust technique for the different variations in human actions
2014 [65]	Ming Zeng et al.	Accelerometer	Discriminative-Scale invariant	Identifying features that clearly separate between activities	Weight sharing technique (partial weight sharing) is applied

Table 1 (continued)

Year	Author	Input source	Feature	Description	Algorithm based technique
2022 [66]	U. Rachna et al.	CCTV	Temporal	To reduce load on hybrid neural network	Dynamic time warping is applied for feature extraction
2007 [51]	Md. Zia Uddin et al.	Depth videos	Spatiotemporal depth silhouette	To obtain the superior recognition and identify abnormal activity	By decoding time-sequential continuous information and generate alarm
2018 [67]	Godwin Ogbuabor et al.	Accelerometer, gyroscope	Time–frequency domain	To improve the detection using ANN	Accelerometer and gyroscope sensor are combined using ANN
2017 [50]	An-Chao Tsai et al.	RGB-D Kinect sensor	Skeleton joints	To enhance visual system of home robot and robot applications	To take advantage of representative 3-D poses data and classify the sequential poses into activity type
2021 [68]	Saeed Mohsen et al.	Smartphone	Time–frequency domain	To improve the classification accuracy up to bar	Precise tunning of algorithm parameters for improvement
2021 [54]	Kemilly Dearo Garcia et al.	Smartphone	Mean, std. deviation, Pearson correlation	To have complex representation of data	To apply semi-supervised learning approach and implement classifier flexible to other models
2020 [69]	Cuong Pham et al.	E-shoes and smartwatch	Global features	Real-time implementation and improvement in performance computation	Improved performance in terms of evaluation metrics
2022 [25]	Rajshri C. Mahajan et al.	CCTV footage	Hand-crafted	To deal with environment variation and actors performing task	Transfer learning is applied to save model training time

the generalization abilities of models to be assessed by testing their performance on unseen data [72]. HAR data sets allow researchers to test and validate their algorithms under various scenarios and conditions. Data set shows significant impact on the performance of algorithm. There are several publicly available data sets for HAR, but specifically for the field of security and surveillance, few data sets are available. Vision based and sensor based are the two main categories of the HAR data set. The vision data have two types RGB data and skeleton data. The sensor data have four types derived from type of sensor used, namely wearable, object, environmental and hybrid sensors. The list of public data sets for HAR along with their specification is mentioned in Table 2. The type of data set is mentioned as either vision-based (V) or sensor-based (S). The specification of data set is provided in form of the type of input data, number of activities (classes), number of users, sampling rate or frame rate, and activities performed.

The activities performed are classified as normal activities, household activities, sports activities and others. These activities are categorized as follows:

- Normal activities: walking, standing, sitting, jogging, etc.
- Household activities: drink from a bottle, answer phone, clap, tight lace, preparing a recipe, cleaning the kitchen, etc.
- Sports activities like jump, gallop sideways, one hand wave, kicking, weightlifting, diving, skating, running, swinging a baseball bat, etc.
- Others: Realistic YouTube videos, surveillance videos, human action extracted from movies.

By defining what constitutes "normal" activities (walking, standing, sitting, jogging), security systems can be trained to detect anomalies or unusual behaviours that deviate from these patterns. Understanding and categorizing household and sports activities can aid in creating more intelligent security systems that are context-aware. By understanding the context and categorizing activities accurately, security systems can reduce false positives.

Referring to Table 2, following inferences can be drawn.

- The most popular data sets are WISDM, UCI Har, UCF, KTH, Weizmann, Kinect, HMDB-51, CAD-60, Opportunity.
- UCF Sports is challenging data set as has many variations like camera motion, view angle, changing background. UCF Sports [48] and daily sports data set is used for analysing sports.
- Weizmann data set has videos of low resolution.
- Hollywood data set is realistic in nature as it contains video sequences taken from Hollywood movies.
- The mentioned data sets contribute significantly to the development ML and DL models capable of recognizing a wide range of human activities. This capability is crucial for enhancing security surveillance systems, enabling them to automatically detect, categorize, and respond to potential security threats with greater accuracy and efficiency
- The UCF crime data set is large-scale video data set having 13 different classes of crime such as arrest, abuse, robbery, and fighting. It has real-world surveillance footage of total 128 h of videos.
- ShanghaiTach, AIRTlab are public data sets available for identifying abnormal events and classifying violence activities, respectively. They might be useful for security and surveillance.
- The other data sets listed in the table can be used to train model and test them on UCF Crime, ShanghaiTech, and AIRTlab to ensure model robustness and performance.

Table 2 Available data sets and their specifications

Data sets	Type	No. of activities	No. of users	Sampling rate/frame rate	Activities performed
WISDM	S	6	51	20 Hz	Normal activities
UCF Crime	V	13	–	–	Crime-related activities, including robbery, burglary
UCI Har	S	6	30	10,299 samples	Normal activities
UCF-101	V	101	25 groups	25fps Resolution: 320 × 240	Sports activities
UCF-50	V	50	25 groups	–	Realistic videos are taken from YouTube
Kinect	V	18	10	30 Hz, 640 × 480	Normal activities
Opportunity	S	5	4	30 Hz	Normal activities
Opportunity++	S + V	20	4	640 × 480, 10fps	Daily Living activities
MSR activity 3D	V	16	10	20 samples	Household activities
HAPT	S	6	30	1214 samples	Normal activities
UniMiB-SHAR	S	17	30	50 Hz	Household activities
Florence 3D	V	9	10	215 samples	Household activities
PAPMP2	S	18	9	100 Hz	Household activities
NTU RGB-D	V	60	40	–	Household activities
Actitracker	S	6	36	20 Hz	Normal activities
Skoda	S	46	–	96 Hz	Assembly line activities
WARD	S	13	20	–	Normal activities
USC-HAD	S	12	14	–	Motion node accelerometer
Daphnet	S	2	10	–	Freeze, no freeze

Table 2 (continued)

Data sets	Type	No. of activities	No. of users	Sampling rate/frame rate	Activities performed
Hollywood	V	8	–	54 samples	Normal activities
Hollywood2	V	12	–	54 samples	Human action extracted from 69 movies
Weizmann	V	10	9		Sports activities
SBHAR	S	12	30	–	Normal activities
HDBD	S	–	28	50 Hz	Normal activities
Egocentric	V	20	10	15 fps Resolution: 320 × 480	Household activities
UCF Sports	V	150	–	20 samples resolution-720 × 480	Sports activities
UCF-ARG	V	5	12	60fps Resolution:1920*1080	Normal activities
Gomaa-1	S	14	3	50 Hz, 603 samples	Normal activities
REALDISP	S	33	17	50 Hz 4000 samples	Normal activities
Daily Sports	S	19	8	25 Hz, 9120 samples	Sports activities
HAD-AW	S	32	16	50 Hz, 4344 samples	
KTH	V	6	25	25 fps Resolution 160*120,	Sports activities
INRIA XMAS	V	14	11	–	Watch checking activities
YouTube action data set	V	11	25	–	Sports activities
i3DPost	V	13	8	–	Sports activities
MOBISERV-AIIA	V	5	12	Resolution—640*480	“having a meal” activities
Gomaa-1	S	14	3	50 Hz Resolution 160*120	Household activities
SPHAR	V	14	–	7759 videos	Realistic actions conducted in real life by actors
CMU-MMAC	S + V	5	18	1024 × 768, 640 × 480	Cooking recipes
LARa	S + V	8	14	–	Normal activities
Berkeley-MHAD	S + V	11	12	–	Normal activities

In Table 2, entries for number of users and sampling rate/frame rate are marked with '-' for some data sets indicates-no specific value was emphasized and hence not mentioned.

The data set is essential parameter for HAR in security and surveillance. The algorithms are performed on the data set in a simulation environment and it gives reasonable performance. These sensors function reasonably well in a simulation setting. However, they could generate a large number of false alarms when employed in realistic scenarios.

In the security domain, image classification faces some challenges like data imbalance, multi-modality, intra-class variation and inter-class similarity and domain adaptiveness.

1. Data imbalance: The raw data recorded from unconstrained conditions are naturally class-imbalanced. When using an imbalanced data set, conventional models tend to predict the class with the majority number of training samples while ignoring the class with few available training samples. Therefore, it is urgent to determine the class imbalance issue for developing an effective activity recognition model.
2. Multi-modality: Security applications often involve data from multiple sources or modalities, such as combining video, audio, and sensor data. Traditional image classification models designed for single-modality data may struggle to effectively utilize information from diverse sources. Integrating features from various modalities can provide a more comprehensive understanding of security scenarios.
3. Intra-class variation and inter-class similarity: Intra-class similarity means two activities of different classes fall into same category. For example, the walking gets misclassified as running because of high speed. Inter-class similarity means if two subjects are performing same activity, it can be bit different because of movements, variation of subject. The same activity falls into two different classes. Jogging activity performed by different subject can get classified as running.
4. Domain adaptiveness: Security environments are dynamic, and the characteristics of data may vary across different domains or scenarios. Achieving domain adaptability is crucial for deploying image classification models in diverse security settings. Methods for domain adaptation, transfer learning, or continual learning are essential to ensure that models can adapt to changes in the environment without a significant drop in performance.

3.1 Representation of vision and sensor-based data set with advantages and disadvantages

In order to understand the pros and cons of security and surveillance, HAR approaches are categorized into sensor and vision-based source of input and corresponding advantages and disadvantages are presented in Fig. 5.

Advantages are shown in green block, while disadvantages are shown in pink block. RGB data and Skeleton data are gathered from CCTV cameras and RGB depth cameras, respectively. Wearable sensors [71] such as an accelerometer, gyroscope, magnetometer, and proximity sensor are included in the sensor data set [72]. RFID and Wi-Fi are both object sensors.

Following are the observations derived from Fig. 5

- CCTV and RGB depth camera provide high accuracy but both are expensive.
- Wearable sensors are cheap, wearable and lightweight. They are more sensitive to environmental conditions.
- Object sensors are having less cost, but they require more resources and get affected by temperature and humidity.

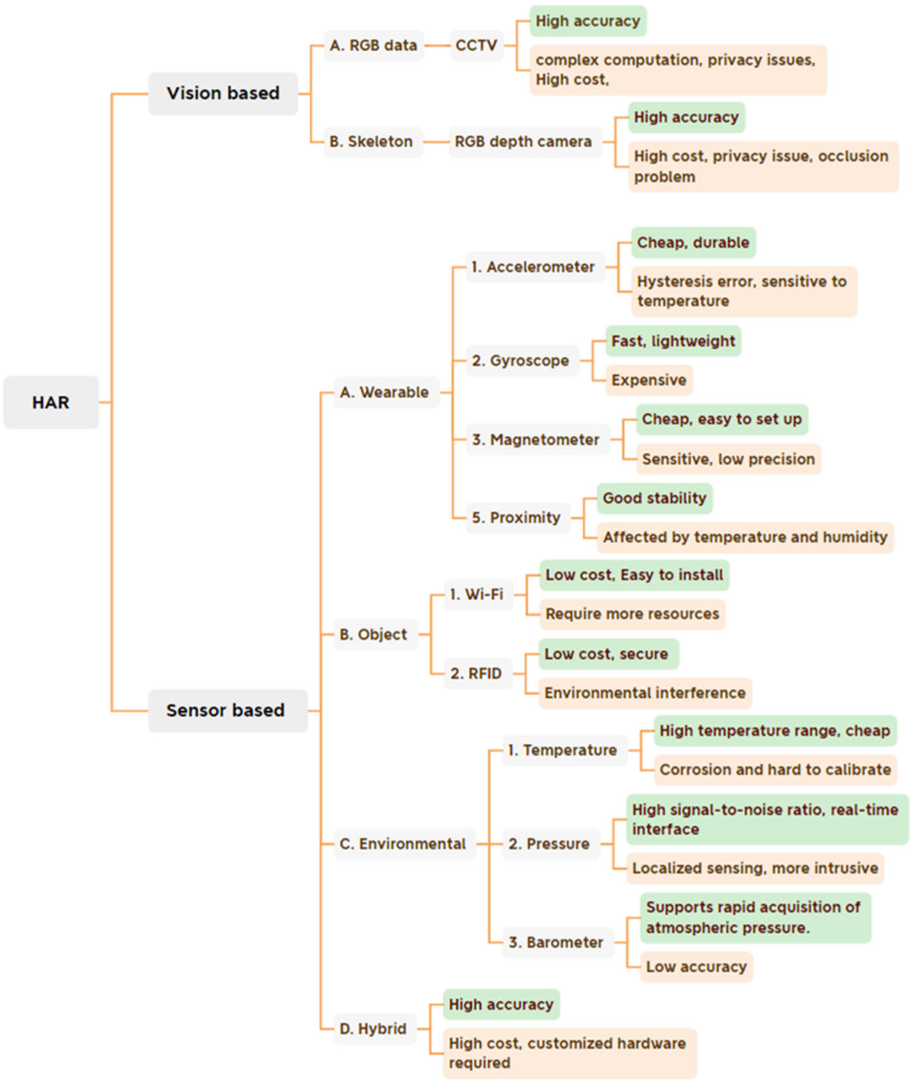

Fig. 5 HAR approaches and its advantages, disadvantages

- Environmental sensors make utilization of environmental characteristics.
- Hybrid sensors give high accuracy, since they are costly and customized hardware is necessary.

The multimodal fusion of sensor and vision information enhances surveillance and security system. By integrating sensor data and video analytics, security systems may gain a more comprehensive understanding of the surroundings, making it easier to discriminate between normal and suspicious activities. This integration can also help to reduce false alarms because the combination of video and sensor data offers a more complete context for evaluating alerts. An approach [73] of feature-level fusion for robust HAR, which utilizes data from multiple sensors, including RGB camera, depth sensor, and wearable inertial sensors can be applicable in surveillance application.

Data set and ML-DL algorithms are the prime components in the implementation of HAR system. These are examined in the Sect. 3. To evaluate the performance of model, there should be understanding of workflow that properly connects the data set and algorithm. A workflow ensures that proper assessment techniques are employed, such as partitioning the data set into training and testing sets, and using appropriate evaluation metrics.

4 Flowgraph of HAR

Referring to literature, combined flowgraph is prepared to indicate important steps doe vision and sensor-based inputs starting from pre-processing to activity classification. Workflow in HAR for security and surveillance ensures that the deployment of reliable and accurate activity recognition systems is there in a methodical and effective manner [74]. Figure 6 gives the detailed flow diagram of HAR for vision and sensor-based data sets. It shows step-by-step and structured approach for performing activity recognition from top to bottom. The raw data collected are firstly spilt into train and test data. According to the data set type either sensor or vision, data pre-processing techniques will be applied. The flowgraph is segmented into sections—data set collection, data set split, pre-processing, feature extraction and model training based on the activities that is carried out.

4.1 Data set collection

Collect the data set containing human activities. It can be labelled or unlabelled. This data set could include sensor data from accelerometers, gyroscopes, other sources or vision data from camera, CCTV.

4.2 Pre-processing

Various filters are applied for noise removal, such as low-pass filter (LPF), mean, median, Butterworth, Kalman filter. The performance factors of these filters such as signal-to-noise ratio (SNR), correlation coefficient, cut-off frequency, order of filter must be taken into account during the filter selection. For vision-based data, pre-processing is performed by video to frame conversion. The frame is detected using different object detection algorithms to show the bounding box. Unnecessary frames can be removed. Noise removal is done with the help of foreground and background subtraction. Normalization, scaling and enhancement are performed using various image processing algorithms. Pre-processing includes important tasks like segmentation and data transformation.

4.2.1 Segmentation

Window length is important for the segmentation task. There is a no specific rule for selecting window length. In previous studies [62], window size range is 0.25–6.7 s. The feature extraction for small window length gives better results but it requires more time. Conversely, activity is performed for a small window duration, then selection of large window can give poor results. Optimal window length is good.

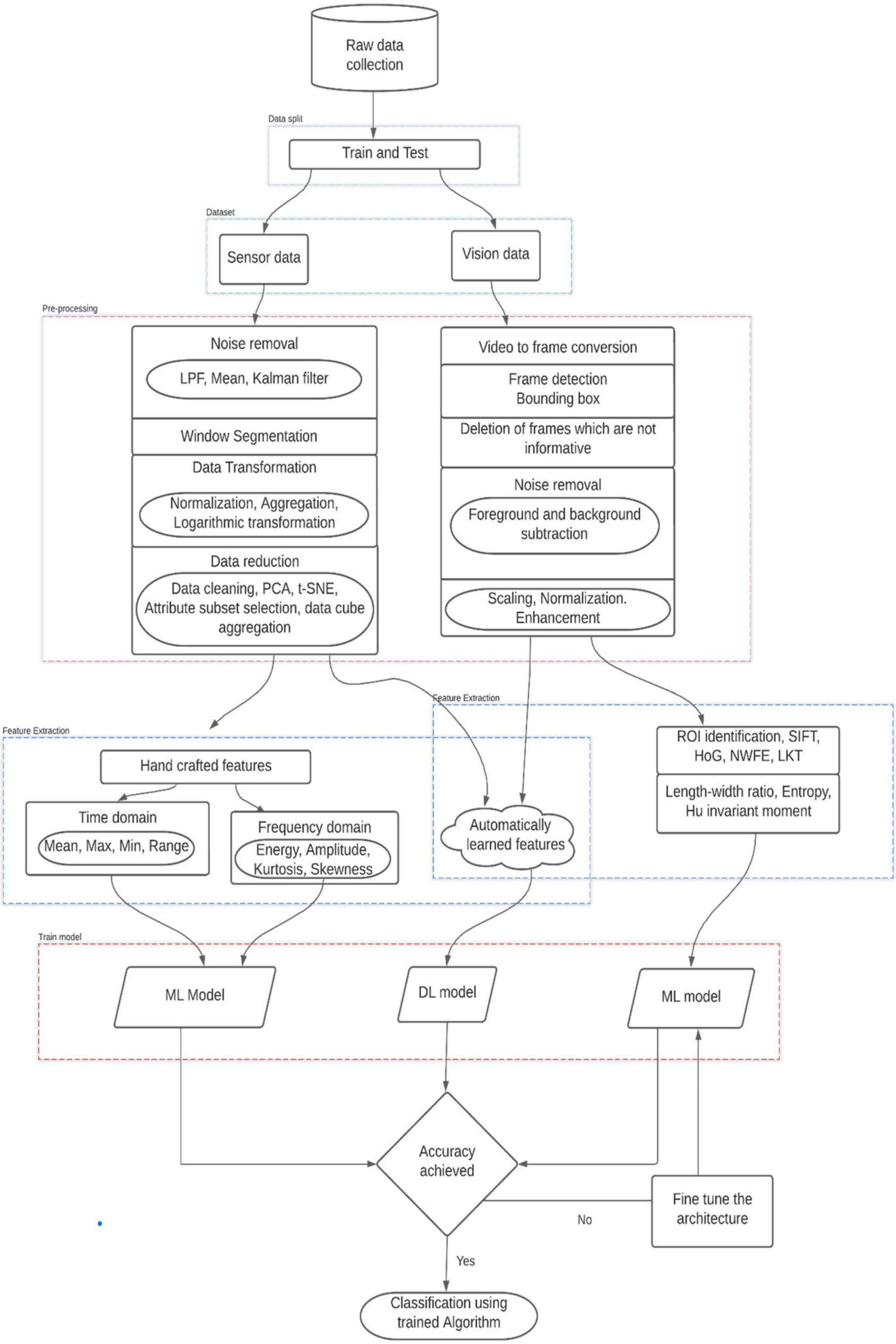

Fig. 6 Flowgraph of HAR

4.3 Split data set

Create training and testing sets from the data set. The test set is utilized for evaluation and performance assessment, whereas the training set is used to train the ML-DL models. The train test split ratio can be different like 80:20, 70:30, etc. depending on size of data set and application.

4.4 Feature Extraction

Feature extraction is key major step in human activity recognition. The features are categorized into two groups—handcrafted features and automatically extracted features. The handcrafted features are manually extracted features which require expert knowledge. In earlier studies handcrafted features were used with traditional ML algorithms. Recently, DL algorithms like CNN extract the features automatically by learning the features [54]. Although DL algorithms have ability to extract features automatically, still we cannot ignore the handcrafted features. Time domain and frequency domain are the two major types of handcrafted features. Features used to separate static features and frequency domain features are used to differentiate between static and dynamic features. The time domain features are mean value, maximum value, minimum value, range, standard deviation, etc. The frequency domain features are energy, power, amplitude, kurtosis, skewness, etc. In case of vision data feature extraction is region of interest (ROI) identification, non-parametric weighted feature extraction (NWFE) based on extension of scatter matrices. It is good for high-dimensional multi-class activity recognition problems. Lucas–Kanade–Tomasi (LKT) is feature tracker. It performs direct search for the position by using spatial intensity information to provide best match. SIFT detects describe and match local features in image. HOG basically utilized for object detection. It takes into account the number of occurrences of gradient orientation in area of image. The simple hand-crafted features for image are edges, corners that can be detected by edge detector algorithms.

4.4.1 Data transformation and reduction

The data need to be transformed by using different transformation techniques so that the data should be in proper format and next operations can be performed smoothly.

The raw data is huge quantity most of the times having various features presented in it. Training such a big data takes lot of time and also it requires complex computational platform. Data reduction methods helps to reduce the dimensionality of data either by combining some features or by removing the less important features. Principal component analysis is dimensionality reduction technique based on eigen values and eigen vectors. T-distributed stochastic neighbour embedding (t-SNE) is nonlinear method which preserves small pairwise distance. Some other techniques are attribute subset selection and data cube aggregation.

4.5 Model training

Model is trained using extracted features for classification of activities into the correct category. ML or DL model can be utilized for training purpose [82]. Model training includes various parameters such as learning rate, hyperparameter tuning, number of epochs, batch size, number of layers, size of kernel, size of filter, activation functions, batch- normalization, optimization algorithms and all. Different optimization algorithms are available in literature.

Gradient descent (GD) optimization is old optimization algorithm. Different version of GD optimization is there such as Stochastic GD, Mini-batch GD, RMS prop, Adagrad [7117]. Recently, Adam is most popular optimization technique applied to models to obtain good accuracy. The trained model is then checked for the required accuracy performance parameter. If the satisfactory results not achieved then find tuning of ML or DL model is performed. The trained model with good accuracy score can be applied to perform other classification tasks. The workflow of HAR for security and surveillance is discussed in the Sect. 4. For the identification and classification of any activity, different phases have to be passed through such as data extraction, data pre-processing, data cleaning, data segmentation, data transformation, feature extraction, feature selection and then train the model. The effectiveness of model training is assessed by utilizing various performance metrics, including accuracy, F1 score, precision, recall, specificity, and AUC to determine the model's performance and suitability for the given task. The important performance parameters for HAR in security and surveillance are discussed and explained in the Sect. 4.6.

4.6 Performance evaluation measures

Most common performance evaluation metrics [74] that are pertinent to HAR in security and surveillance are discussed in Table 3.

The analysis is made by evaluating some review papers having different 35 data sets to find out most popular performance metrics used in security and surveillance area. Referring to the review data, performance measures impact is illustrated in Fig. 7., while it can be observed that **Accuracy** and **F-1 score** parameters are most commonly used. The accuracy parameter quantifies the percentage of correctly identified activities out of all instances. A high detection accuracy shows that system can reliably recognize the relevant activities and contributes to improving system security. However, accuracy metric is beneficial for balanced data set, while F-1 score metric is good in case of imbalanced data set as it takes into account the combination of both precision as well as recall parameters. The public data sets available for human activity recognition in security and surveillance domain is imbalanced in most of the cases. The next important parameter is **Precision** that indicates the prediction ability of individual category. Precision can be applied to security and surveillance related HAR. It measures the proportion of correctly identified threats out of all detected activities. **Recall** parameter indicates the proportion of true suspicious activities correctly identified. A good balance between precision and recall is important to minimize false positive and false negative of the security and surveillance system. **FPR** gives occurrence of the false positive detection in form of number, where system identifies the activity as suspicious but it is not. It makes sure that system is genuine and focus on its objectives rather than false alarm, also known as False alarm rate. To evaluate the entire performance of the HAR system in security and surveillance applications, it is crucial to take these performance measures into account collectively. Since it can be observed that the accuracy parameter is considered in the majority of situations, as per referred literature a Table 4 is supplied to indicate the accuracy value produced by employing various ML-DL algorithms. But along with that, real-time detection and speed are two critical factors required for security and surveillance. Different approaches applied to the same data set result in varying accuracy parameter values. This is due to the diversity of the algorithm and its architecture.

It is observed that the accuracy of a HAR system using the same ML algorithm for different data sets can vary depending upon several factors like different data size, different data distribution, different feature representation. As shown in Table 4, HMM gives different

Table 3 Performance evaluation parameters

Performance Measures	Definition	Formula
Accuracy	Correct predictions calculated over total predictions	$\frac{TP+TN}{TP+TN+FP+FN}$
F-1 Score	Harmonic mean of recall and precision	$2*\frac{Precision*Recall}{Precision+Recall}$
Precision	Positive predictive value that reflects how reliable the model is in classifying samples as positive	$\frac{TP}{TP+FP}$
Recall/ Sensitivity/TPR	Measures the model's ability to detect positive samples	$\frac{TP}{TP+FN}$
Specificity	Measures the proportion of correctly identified negative occurrences to all true negative occurrences	$\frac{TN}{TN+FP}$
FPR	Ratio between total expected negative observations and false positive observations	$\frac{FP}{FP+TN}$
AUC	Presents the likelihood of the model distinguishing observations from two classes	$\int_0^1 TPR(FPR^{-1}(x))dx$
Log-Loss	It considers uncertainty of prediction	$\frac{-1}{N}\sum_{i=1}^{N} yi \cdot \log(p(yi)) + (1-yi)\cdot \log(1-p(yi))$
Kappa	Compares the observed accuracy to an expected accuracy or the accuracy expected from random chance	$\frac{(Observed\ Accuracy-Expected\ Accuracy)}{1-Expected\ Accuracy}$
Likelihood ratio	The ratio of the likelihood that a particular activity would be predicted when it matches the ground truth to the likelihood that it would be predicted erroneously	$\frac{sensitivity}{1-specificity} = \frac{TP(TN+FP)}{FP(TP+FN)}$
Mean Average Precision	Each class has its own accuracy and recall values, and the average precision is obtained by averaging the precision values at various recall levels	$\frac{\sum_{q=1}^{Q} AveP(q)}{Q}$

FP False Positive, *TP* True Positive, *FN* False Negative, *FP* False Positive, *FPR* False Positive Rate, *TPR* True Positive Rate

accuracy values for MSR daily activity 3D (94.4) and Florence 3D (96.2). DL algorithms, especially CNN and RNNs like LSTM, Bi-LSTM, have shown significant improvements in HAR accuracy. Some methods combine the two or more algorithms like CNN + LSTM, SIFT + SVM, etc. to obtain the modified result. Hybrid algorithm like CNN + LSTM combine spatial feature extraction from CNNs with temporal modelling and sequential learning from LSTMs like 92% on UCI-HAR and 95.87% on Open HAPT data set. This hybrid algorithm shows 83.13% accuracy on UMN data set but after the adding an attention mechanism, accuracy increases and reaches to 94.30%.

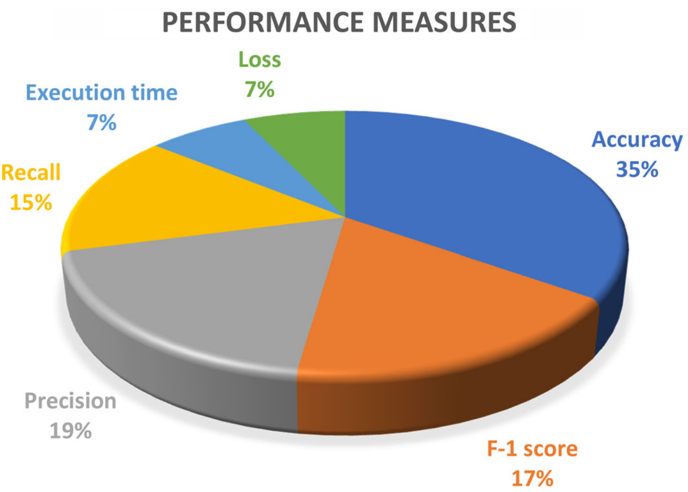

Fig. 7 Performance measures used in HAR

Table 4 Performance comparison for human activity recognition

Type	Data set	Architecture of algorithm	Accuracy	Reference
Sensor data	WISDM	SAE	85.36%	[74]
	WISDM	Ensemble of AE	89.32%	[53]
	UCI-HAR	KNN	90.46%	[68]
	Uni-MiB-SHAR	HMM, RF & high-level feature	91.1%	[58]
	UCI-HAR	**CNN + LSTM**	92%	[41]
	Opportunity	ConvLSTM	92.2%	[37]
		CNN + LSTM	92.63%	[42]
	MHEALTH	Fusion ResNet	93.4%	[77]
	UMN	CNN + LSTM + Attention	94.30%	[36]
	Open HAPT	**CNN + LSTM**	95.87%	[61]
	CSL-SHARE	HMM, RF & high-level feature	96.1%	[57]
	UCI-HAR	SAE	96.34%	[74]
	iSPL	**CNN + LSTM**	99%	[41]
Vision data	Weizmann	YOLOv3	80.20%	[80]
	KTH	SIFT + SVM	82%	[49]
		CNN	97%	[55]
		VGG-19 and Naïve bayes	97%	[78]
	IXMAS	**VGG-19 and Naïve bayes**	97%	[78]
	UCF Sports	**VGG-19 and Naïve bayes**	98%	[77]
	YouTube	**VGG-19 and Naïve bayes**	99.4%	[77]
	SPHAR	2D-CNN + SPDB LSTM	94.5%	[79]
	ImageNet	DCNN	99.82%	[81]
		DCGAN + DCNN	99.91%	[81]

5 Discussion

In this paper, an extensive review regarding the role of traditional feature descriptors, ML and DL in HAR system for security and surveillance area is conducted by considering the literature available to date. HAR is continuously developing because of number applications in different fields. Current trends and techniques are presented in this paper, as most of the literature reviewed in this review is from the last five years. Different applications of HAR and their contribution to the respective field are identified. This review highlights the broader impact of HAR in security and surveillance, ranging from improving security measures to ensuring public safety. The literature survey may provide specific examples of how HAR has been employed in security and surveillance scenarios, such as recognizing suspicious behaviour or identifying unauthorized access. Figure 2 summarizes the traditional, ML and DL algorithms available for HAR. Various hybrid models presented like CNN + LSTM, SVM + CNN are also discussed. Empirical observations indicate that hybrid models outperform individual modes, demonstrating superior performance in various tasks. The different combination of ML-DL hybrid models can be implemented to improve the performance and reduce the computational power burden for security and surveillance application. Sensor and vision-based data set are the two broad categories of HAR data set. The limitations of sensor and vision-based data are provided in detail. Available data sets for HAR with their specifications are discussed. Analysis based on methodology, input source, accuracy, and data sets are presented. The various performance evaluation metrics available for activity classification are studied and important parameters for security and surveillance application are mentioned with rationale. HAR enables security experts in recognising patterns, enhancing security tactics, and making data driven judgements to strengthen security measures. This paper is valuable resource for researchers, practitioners, and system developers working in this field of security and surveillance as it provides in-depth understanding of cutting-edge techniques and recent advancements in HAR.

6 Conclusion

HAR is now an extensively growing field and numerous applications in different domains. HAR is more critical in security and surveillance than other applications as it deals with public safety, crime prevention, suspicious activity identification, privacy protection. HAR has enormous potential for strengthening security and surveillance systems. The evolution of HAR techniques was examined in this literature review, from early approaches like Haar and LBP to adoption of ML algorithms up to the development of DL models. The Haar and LBP feature descriptors are very complex in nature and provides very less accuracy. ML algorithms use these feature descriptors for feature extraction and then apply the ML models like SVM, RF, DT to classify the activities and gives good results. In case of ML algorithms, with relatively simple activities and limited resources for real-time applications, NB and KNN might be suitable choices. For more complex activities and larger data sets, RF and SVM could provide better accuracy and generalization. The integration of deep learning models has led to the improved the accuracy and robustness of HAR in security and surveillance applications. Some hybrid architectures, such as Convolutional Recurrent Neural Networks (C-RNN), which combine the strengths of CNNs and RNNs provides more improvement in system models.

The selection of ML or DL models depends on the size of a data set, hardware resources, real-time requirement, model interpretability, and computation requirement. Although DL is broader family of machine learning, it does not mean that deep learning is always better than machine learning. ML models are statistical, while DL models are dynamic. The interpretability of DL models is low and it requires more parallel processing unit. If the available data set is small and simple, then apply ML model that will save computational requirement, less time complexity and space complexity, reduces the cost of GPU, TPU. However, for complex activities it is better to choose DL models for classification and recognition. The knowledge gathered from this literature review can open the door for more investigation and the creation of strong HAR systems that strengthen security protocols and promote safer surroundings.

7 Future directions

The continued development of sensor and vision technologies, ML algorithms, and the rising desire for more effective and reliable security systems are the main factors of the future directions for HAR in security and surveillance. Future research might concentrate on creating reliable multi-modal sensor fusion approaches to combine data from many sensors including wearable technology, depth sensors, and video cameras and take advantage of their complimentary capabilities to increase the accuracy and reliability of activity recognition. By combining ML and DL models, activity recognition can be performed in short period and it can be used for real-time application as real-time HAR is crucial for security and surveillance. The use of unsupervised and semi-supervised learning approaches can be investigated in the future to reduce the dependency on massive volumes of labelled data. To facilitate the transfer of information from one domain to another, future research can examine transfer learning and domain adaptation strategies to make HAR systems more flexible and reliable in a variety of security and surveillance applications.

Author contributions All the authors contributed equally in preparing the manuscript.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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